

Introduction to Deep Learning for Life Sciences

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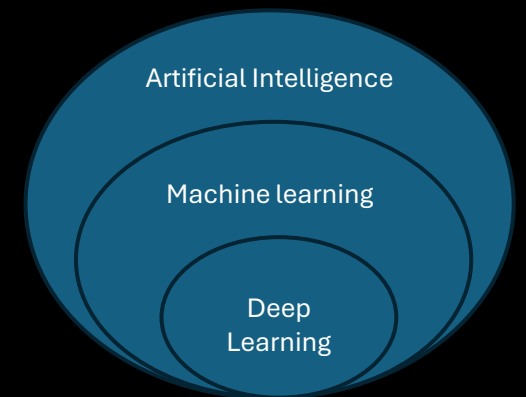
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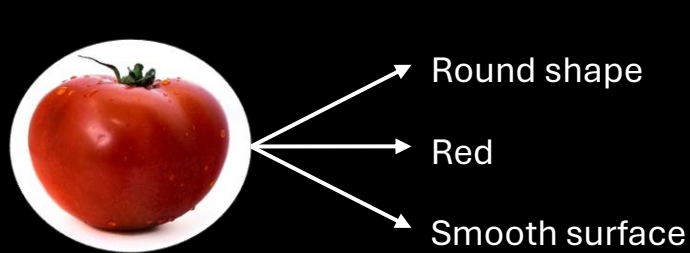
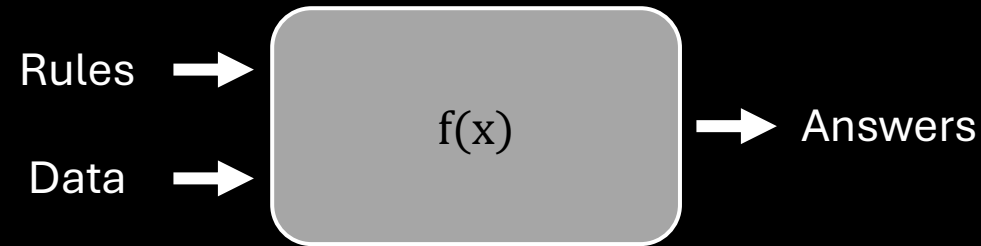
Deep Learning (DL) is a subset of Machine Learning algorithms

- Artificial Intelligence (AI) – term invented in the 1950s: “(...) *the effort to automate intellectual tasks normally performed by humans* (...)”
- Are expected to have some level of “reasoning”
- DL and Artificial Neural Networks have been around for many decades.
 - 1943: *Perceptron*, the “first” artificial neural network, by Warren McCulloch and Walter Pitts
 - 1960s: Deep Neural Networks first appeared in the 1960s
 - 1986: modern backpropagation algorithm by Geoffrey Hinton (Nobel and Turing award laureate)
 - 1980s: Convolution Neural Network (CNN) are used for image recognition (LeNet by Yann LeCun and others)
 - 2010s: start of DL boom (AlexNet and ImageNet challenge)



Shift in computational paradigm

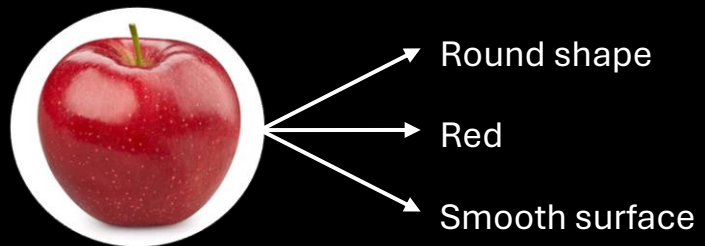
Classical programming



Machine learning method



"tomato"



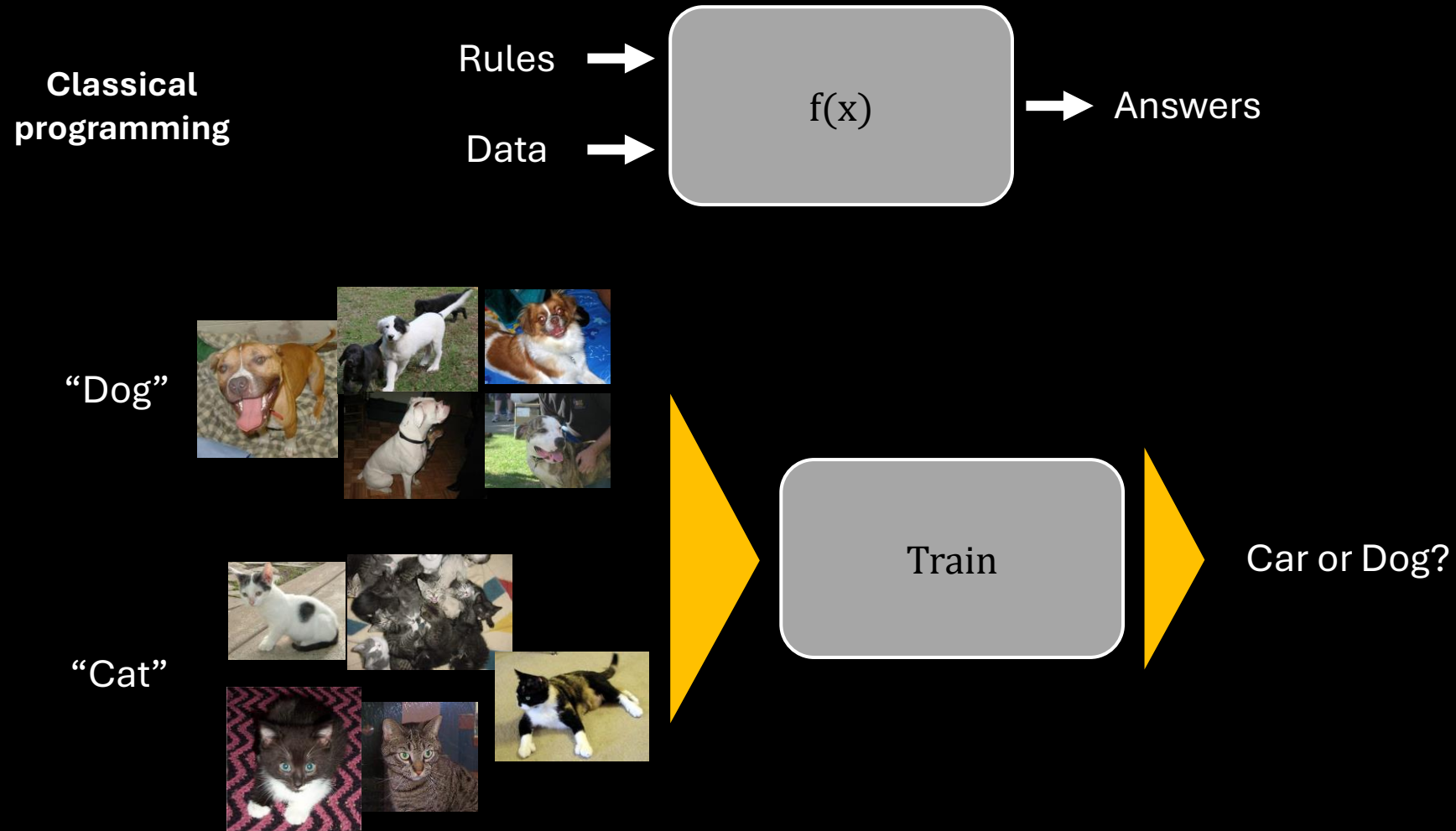
Machine learning method



"tomato"

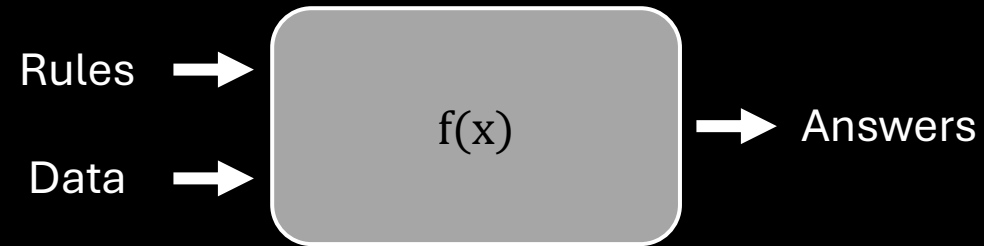
What if the algorithm learns from data automatically?

Shift in computational paradigm

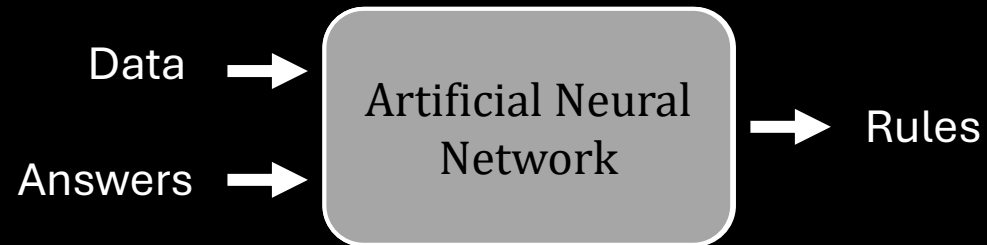


Shift in computational paradigm

**Classical
programming**

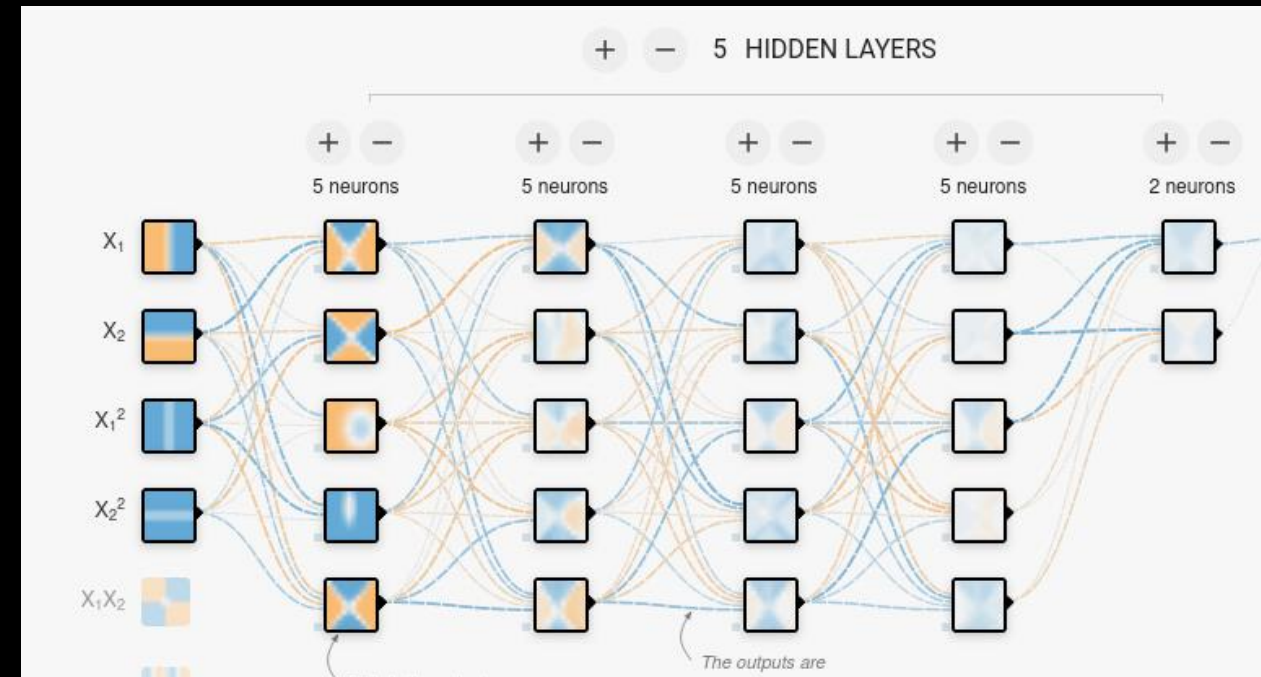


**Deep
Learning**



What are Artificial Neural Networks?

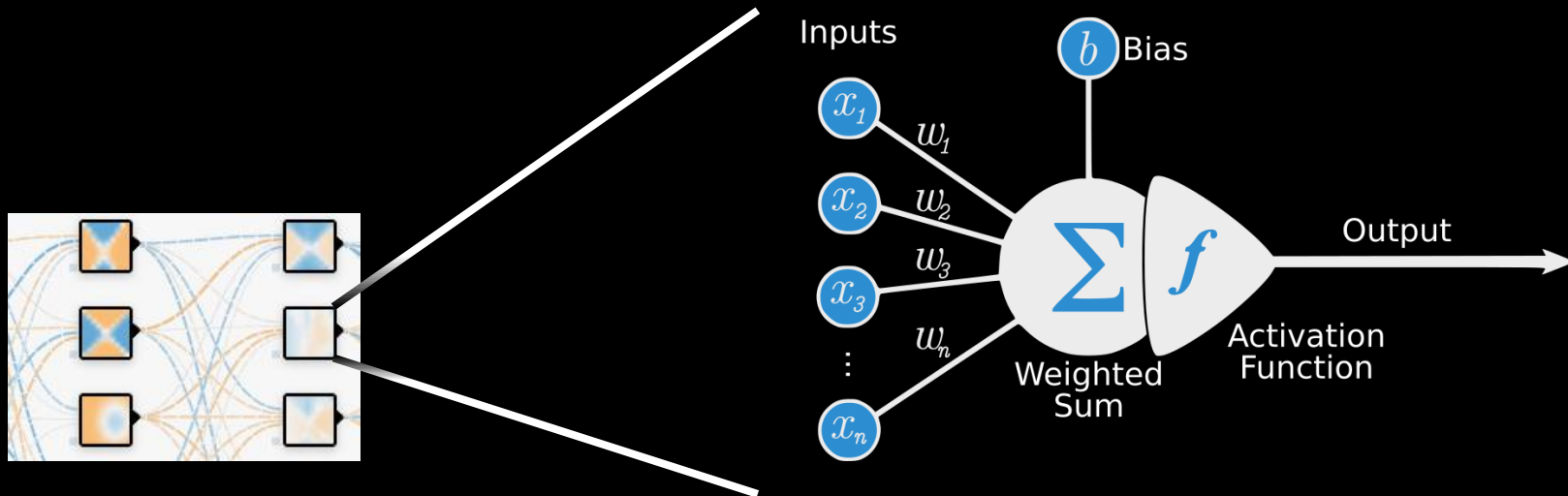
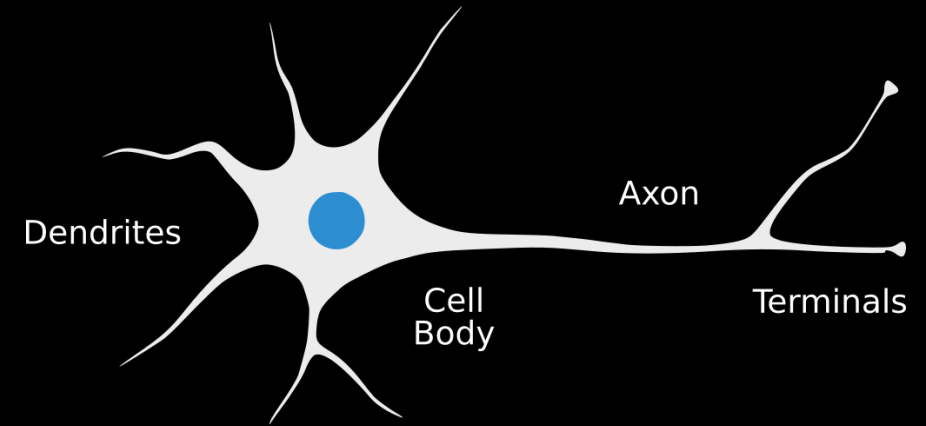
- Artificial Neural Networks (ANN) were inspired by neuronal connections
- Network of connected layers (vary depending on network architecture)
- Each layer is composed by many neurons
 - Nodes on the layer
- Deep Learning networks are characterized by having many “hidden” layers



Tensorflow Playground (<https://playground.tensorflow.org>)

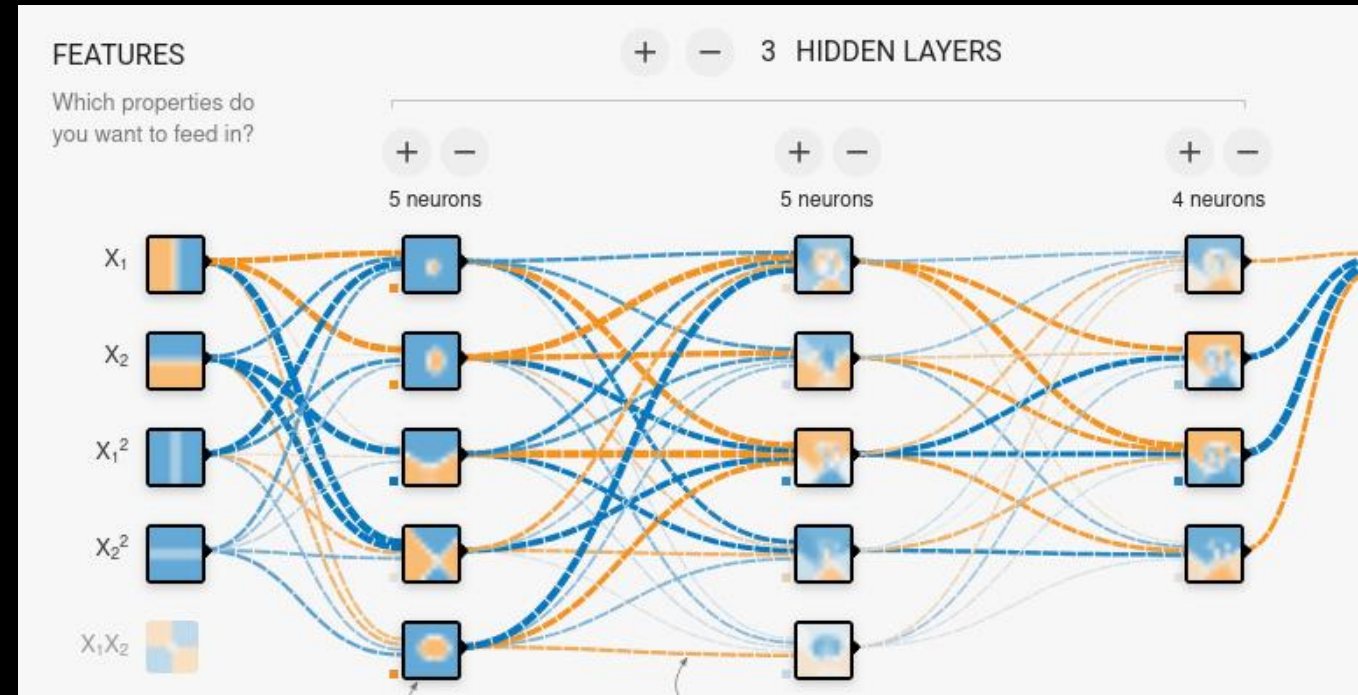
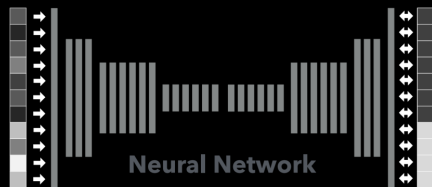
What are Artificial Neural Networks?

- Input from previous nodes are combined and summed (with weights)
- Each network node passes information when activated
 - Signal goes over a certain threshold



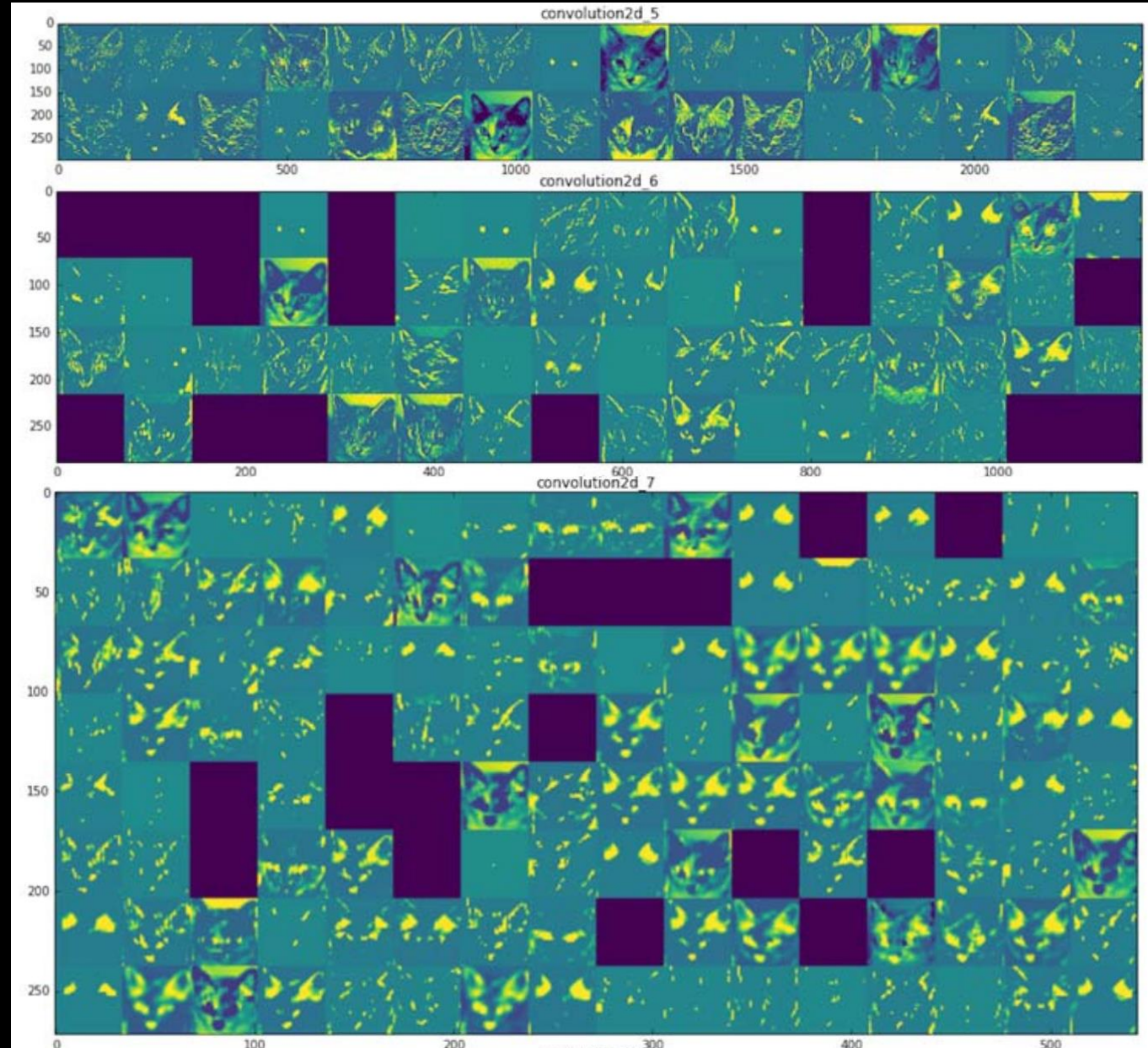
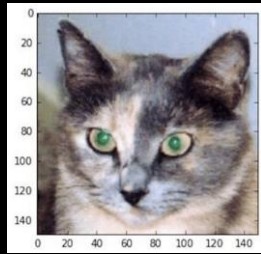
What are Artificial Neural Networks?

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 - Signal goes over a certain threshold
- Different image patterns activate different subsets of neurons
- Most computer vision and image processing ANN are Convolutional Neural Networks (CNN)

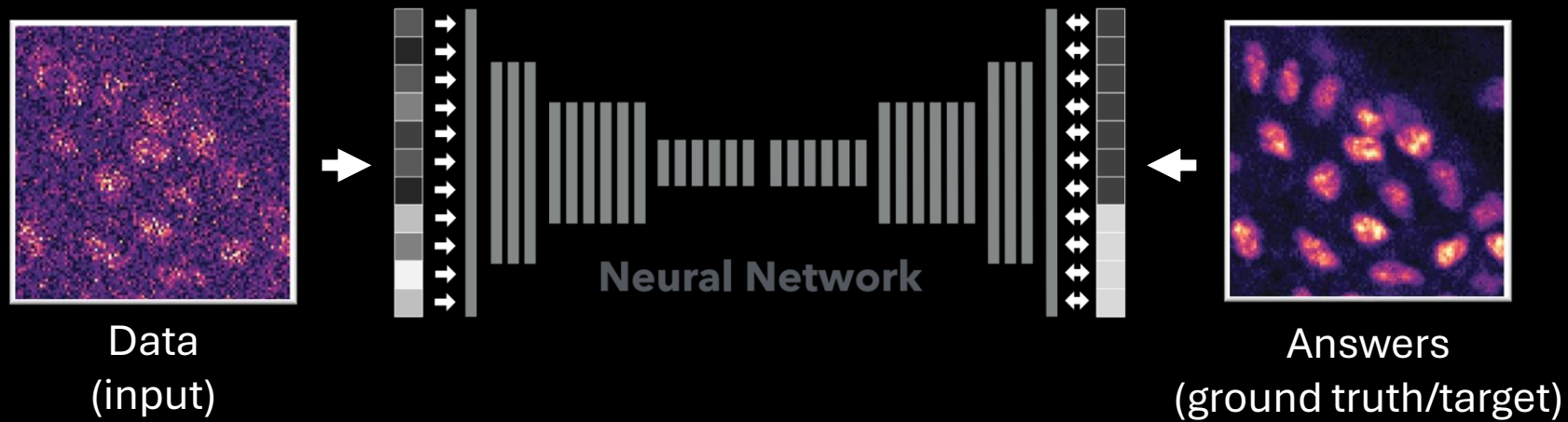


Tensorflow Playground (<https://playground.tensorflow.org>)

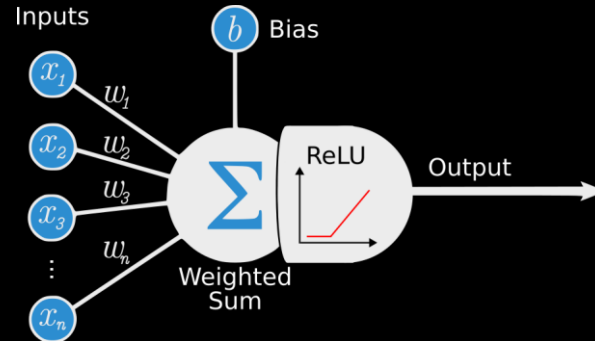
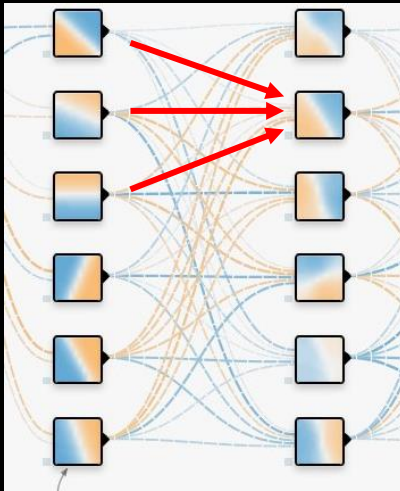
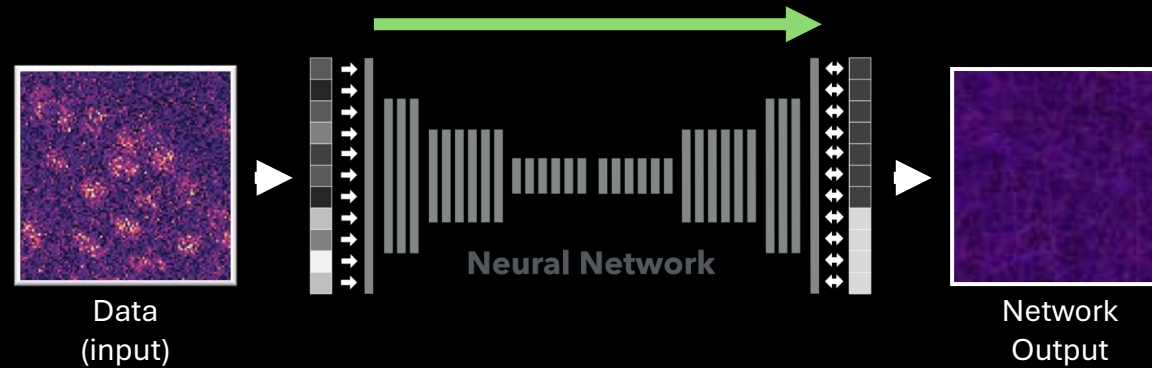
ANN can generate complex filters which are more “abstract”



How do you train a DL network/model?

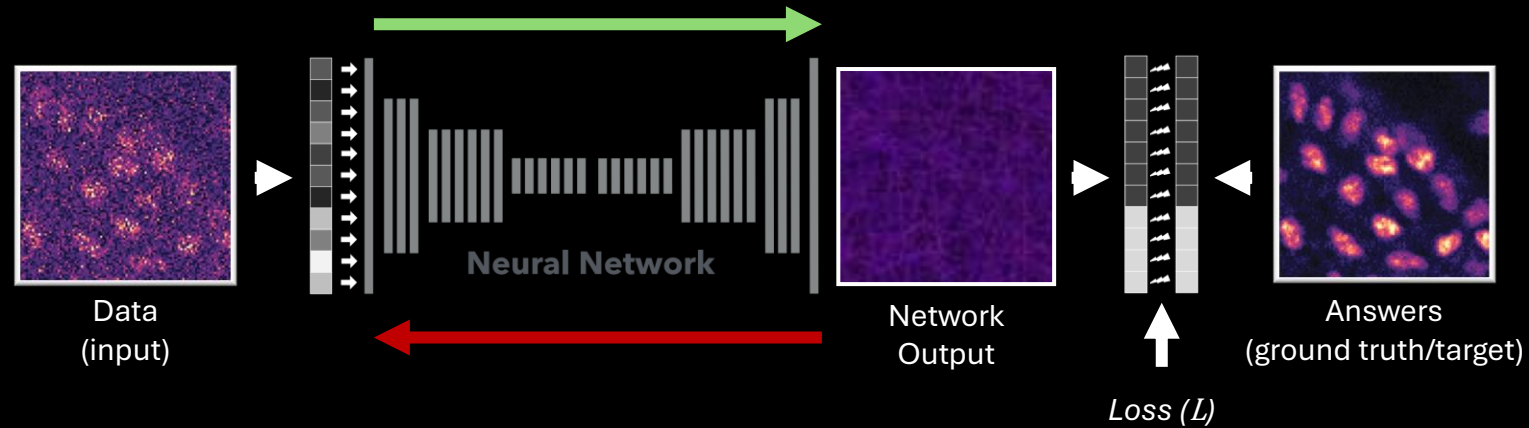


Forward pass sets values and weights with node activations



- Each node input is multiplied by a weight
- All weighted inputs are summed
- Pass through activation function
- If: output > activation threshold → signal passed on
 - ReLU (rectified linear unit) is a common activation function

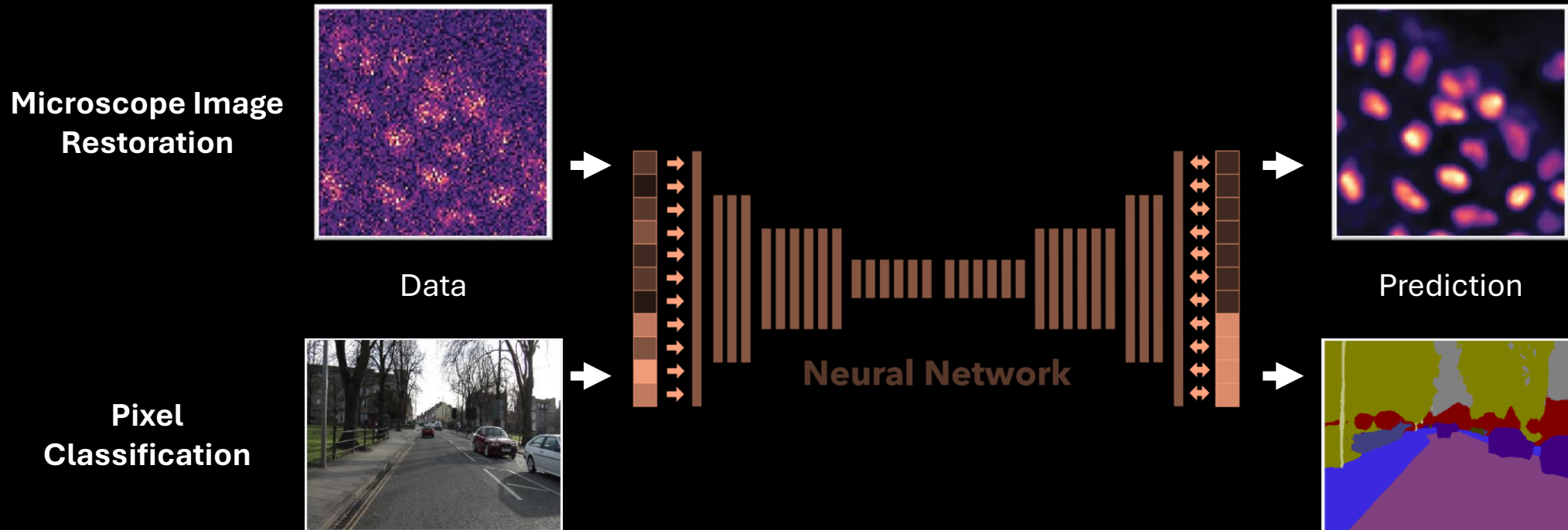
Training loss tells the network how wrong it is and how to adjust



- Network output is compared to ground truth data
- A *Training Loss* (or *error*) is measured
 - Metric that measures difference between output and truth
- Error is backpropagated through the network
- Rinse and repeat (the training part of a network)
 - Over many steps and epochs

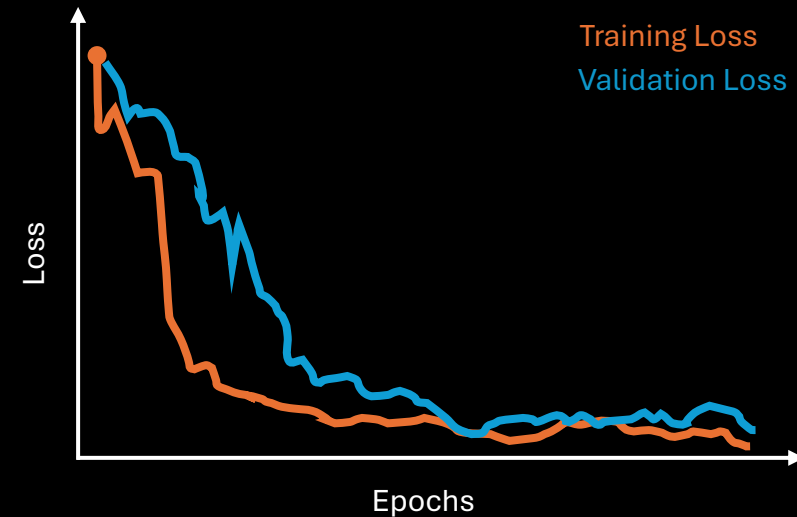
Applying a trained network/model

- Typically called *Inference* or *Prediction*, is the application of a trained model (network) to new unseen data.



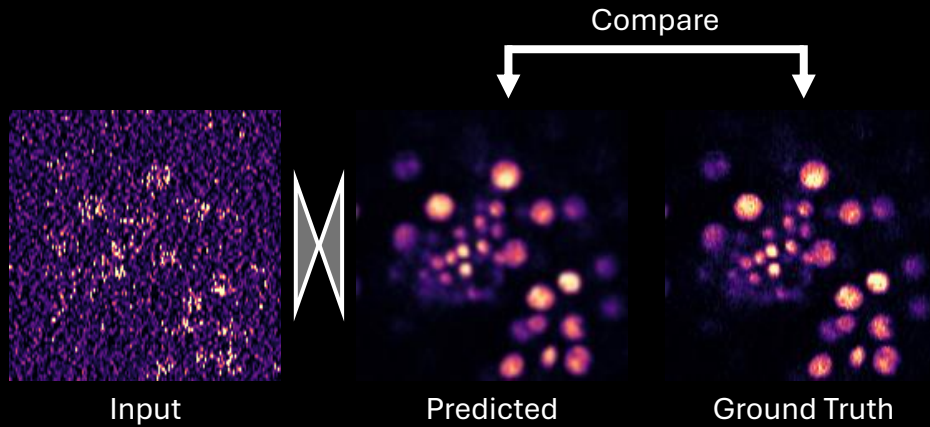
How to follow network training?

- Each forward and backward pass is a step
 - A step passes a batch of images
- An epoch is when the network sees all the training data
 - Many steps until all data is seen
- Training happens over many epochs
 - Training loss converges to zero (0)
- Split data into 3 groups:
 - Training: used to update and tune the network
 - Validation: test the model for unseen data during training
 - Test: new data to test model generalization properties
 - Foundational Models are the best generalizers, trained with huge amounts of data

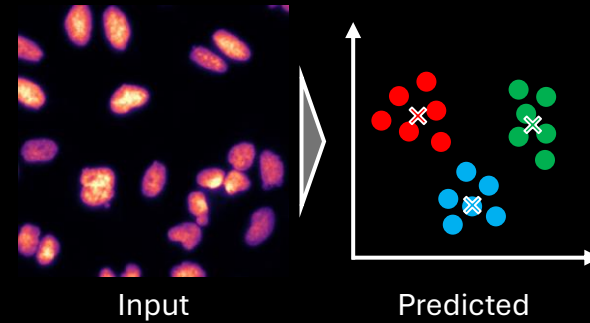


Learning Strategies

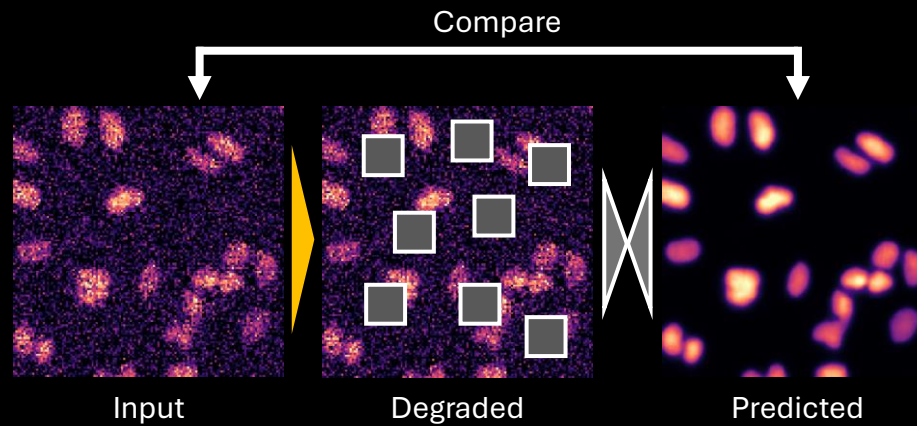
Supervised



Unsupervised

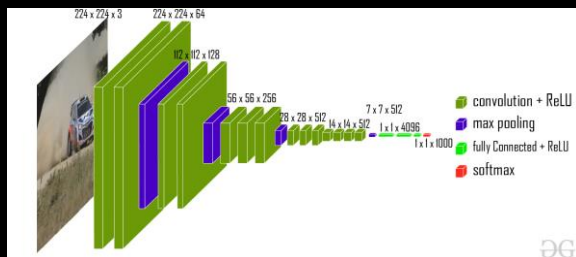


Self-Supervised

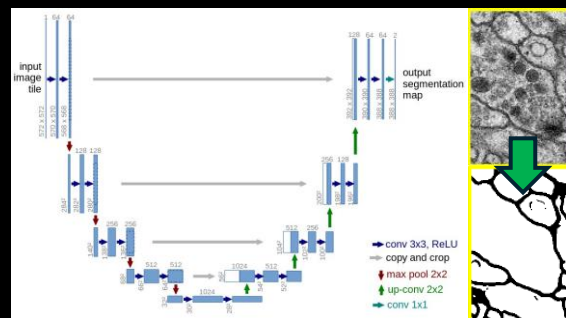


Common DL architectures and their use cases

VGG-16 (CNN)
classification

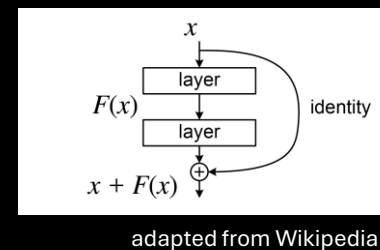


U-Net (CNN)
image segmentation (Classification)

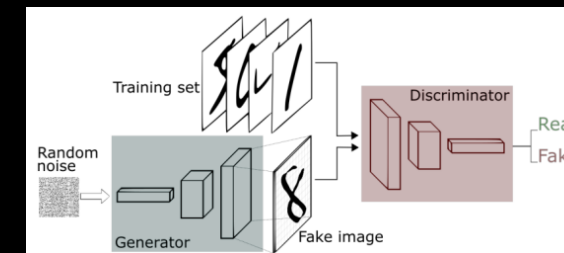


Ronneberger *et al.* 2016

ResNet
image classification

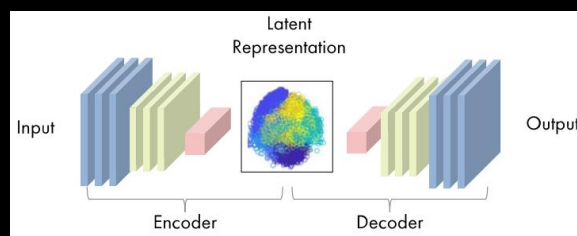


Generative Adversarial Networks (GAN)
image generation

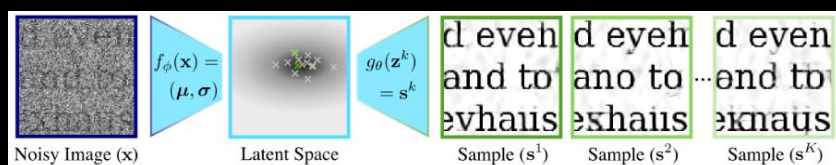


Schematic by Mariana Ferreira

Variational Auto-Encoder
image generation

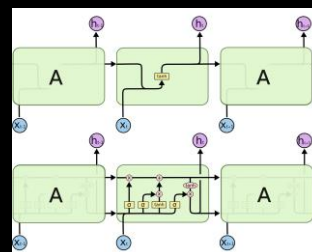


Schematic by Mariana Ferreira



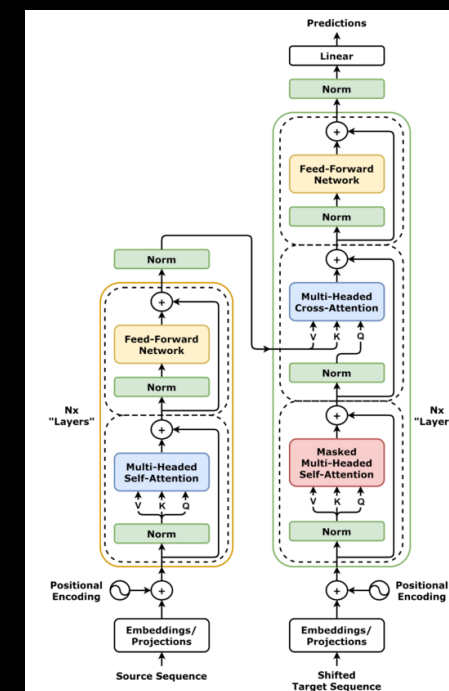
Prakash *et al.* 2021 DivNoising

Long short-term memory
sequential data



Transformers
text/sequential data

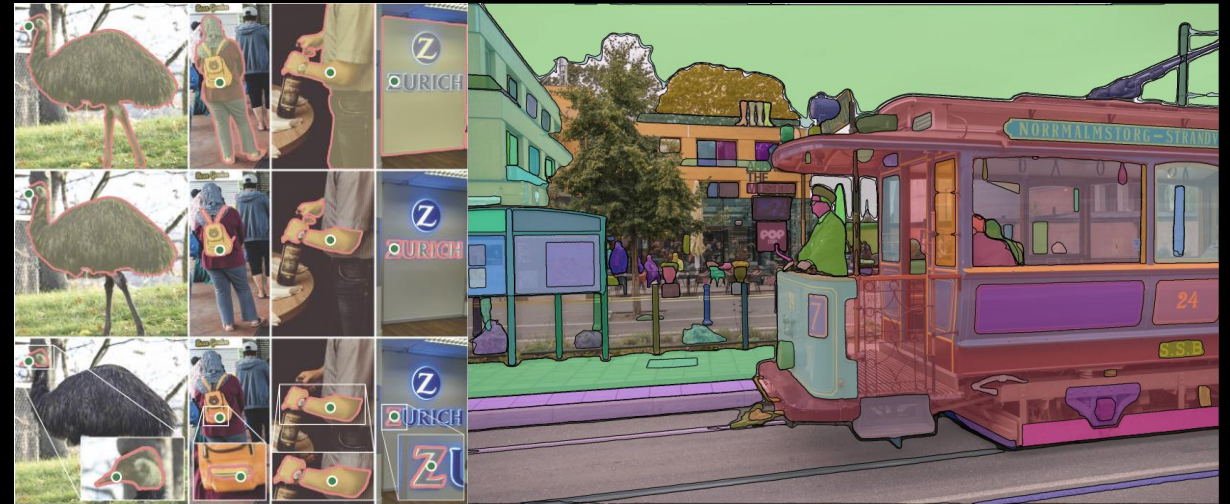
Attention is all you need. *Neurips* 2017



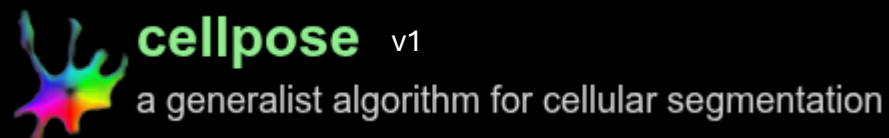
Foundation models

- Large models trained on vast amounts of data (and some can perform general tasks).
 - Large Language Models (LLM) are a type of foundation models
- Google's BERT: baseline for natural language processing
- OpenAI's GPT
 - LLM: ChatGPT, Claude, Llama, Gemini, Grok, etc
- Image generation models:
 - Dall-E (OpenAI)
 - Stable Diffusion (Stability AI)
 - Imagen (Google)

Segment Anything Model (SAM) from Meta AI



adapted from Meta AI (<https://github.com/facebookresearch/segment-anything>)



Cellpose-SAM